

# **SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE**

**SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical**

## **PRACTICAL - 7 (M2)**

**AIM** : Performing one-way ANOVA using aov() (R).

### **Input** :

```
# S101 RAJDEEPP .M. PARAB
# PRACTICAL 7: ONE-WAY ANOVA
# Dataset: diabetess_risk_dataset.csv
# Load the dataset
diabetes_risk <- read.csv("diabetes_risk_dataset.csv")
# Check structure to confirm column names
str(diabetes_risk)
# Convert physical_activity_level to factor
diabetes_risk$physical_activity_level <- as.factor(
  diabetes_risk$physical_activity_level
)
# Perform one-way ANOVA
anova_one_way <- aov(glucose_level ~ physical_activity_level,
  data = diabetes_risk)
# Display ANOVA table
summary(anova_one_way)
# HYPOTHESIS DECISION
p_value <- summary(anova_one_way)[[1]][["Pr(>F)"]][1]
if (p_value < 0.05) {
  cat("Result: Reject the Null Hypothesis\n")
  cat("Reason: Glucose levels significantly differ based on physical activity
level.\n")
} else {
  cat("Result: Accept the Null Hypothesis\n")
  cat("Reason: No significant difference in glucose levels across activity
levels.\n")
}
```

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## Output :

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
R - R4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/
> setwd("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/")
> diabetes_risk_dataset <- read.csv("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/diabetes_risk_dataset.csv")
> View(diabetes_risk_dataset)
> # S101 RAJDEEPP M. PARAB
> # PRACTICAL 7: ONE-WAY ANOVA
> # Dataset: diabetes.csv
> # Load the dataset
> diabetes_risk <- read.csv("diabetes_risk_dataset.csv")
> # Check structure to confirm column names
> str(diabetes_risk)
'data.frame': 100000 obs. of 7 variables:
 $ age          : int  58 71 48 34 62 27 40 58 77 38 ...
 $ bmi          : num  33.2 26.8 21 28 28.3 ...
 $ glucose_level : num  71 126 61.9 137.6 65.9 ...
 $ physical_activity_level: chr  "low" "low" "moderate" "low" ...
 $ family_history : int  0 0 1 0 0 0 1 0 1 1 ...
 $ smoker       : int  0 0 1 0 0 0 1 0 0 0 ...
 $ at_risk_diabetes : int  1 0 0 0 1 0 0 0 0 ...
> # Convert physical_activity_level to factor
> diabetes_risk$physical_activity_level <- as.factor(
+   diabetes_risk$physical_activity_level
+ )
> # Perform one-way ANOVA
> anova_one_way <- aov(glucose_level ~ physical_activity_level,
+   data = diabetes_risk)
> # Display ANOVA table
> summary(anova_one_way)
          Df Sum Sq Mean Sq F value Pr(>F)
physical_activity_level  2      539      269.5    0.354    0.702
Residuals              99997 76222276      762.2
> # HYPOTHESIS DECISION
> p_value <- summary(anova_one_way)[1][["Pr(>F)"]][1]
> if (p_value < 0.05) {
+   cat("Result: Reject the Null Hypothesis\n")
+   cat("Reason: Glucose levels significantly differ based on physical activity level.\n")
+ } else {
+   cat("Result: Accept the Null Hypothesis\n")
+   cat("Reason: No significant difference in glucose levels across activity levels.\n")
+ }
Result: Accept the Null Hypothesis
Reason: No significant difference in glucose levels across activity levels.
>
>
> source("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/Practical 7 (M2).R")
'data.frame': 100000 obs. of 7 variables:
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
R - R4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/
'data.frame': 100000 obs. of 7 variables:
 $ age          : int  58 71 48 34 62 27 40 58 77 38 ...
 $ bmi          : num  33.2 26.8 21 28 28.3 ...
 $ glucose_level : num  71 126 61.9 137.6 65.9 ...
 $ physical_activity_level: chr  "low" "low" "moderate" "low" ...
 $ family_history : int  0 0 1 0 0 0 1 0 1 1 ...
 $ smoker       : int  0 0 1 0 0 0 1 0 0 0 ...
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> anova_one_way <- aov(glucose_level ~ physical_activity_level,
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> # Display ANOVA table
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+   cat("Result: Reject the Null Hypothesis\n")
+   cat("Reason: Glucose levels significantly differ based on physical activity level.\n")
+ } else {
+   cat("Result: Accept the Null Hypothesis\n")
+   cat("Reason: No significant difference in glucose levels across activity levels.\n")
+ }
Result: Accept the Null Hypothesis
Reason: No significant difference in glucose levels across activity levels.
>
>
> source("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/Practical 7 (M2).R")
'data.frame': 100000 obs. of 7 variables:
 $ age          : int  58 71 48 34 62 27 40 58 77 38 ...
 $ bmi          : num  33.2 26.8 21 28 28.3 ...
 $ glucose_level : num  71 126 61.9 137.6 65.9 ...
 $ physical_activity_level: chr  "low" "low" "moderate" "low" ...
 $ family_history : int  0 0 1 0 0 0 1 0 1 1 ...
 $ smoker       : int  0 0 1 0 0 0 1 0 0 0 ...
 $ at_risk_diabetes : int  1 0 0 0 1 0 0 0 0 ...
Result: Accept the Null Hypothesis
Reason: No significant difference in glucose levels across activity levels.
>
>
```